A MICROFLUIDIC DEVICE TO PROMOTE EXOSOME SECRETION

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ABSTRACT

This paper reports a high-throughput microfluidic device to promote exosome secretion from mesenchymal stem cells. Through a microchannel with ridge gap height smaller than cell size, mechanical stimulation acting on the cells when they pass through the squeezing ridges can effectively permeabilize the cell membrane, thus boosting the cells to secrete exosomes into extracellular space. The results show that the secretion amount of exosomes from the cells treated by the microfluidic device increases by 3.1-fold without affecting their bio-activity, thus providing a potential platform to improve exosome yield for biomedical research.

KEYWORDS: microfluidic devices, high throughput, exosomes, secretion amount, mesenchymal stem cells

INTRODUCTION

Exosomes are nanometer-sized extracellular vesicles (40 – 200 nm in size) that are naturally secreted by almost all cell types into extracellular microenvironment, serving as an intercellular signal mediator. Since exosomes play a significant role in various physiological and pathological activities, they have been widely used as promising biomarkers for early diagnosis and natural carriers for targeted drug delivery. However, the practical applications of exosomes are hindered by their trivial secretion amount from donor cells, especially when using mesenchymal stem cells (MSCs) because of their limited expansion capability. To overcome this obstacle, several attempts have been presented, such as acoustic radiation and electrical stimulation acting on bulk samples.[1] However, these methods usually suffer from inconsistent cell stimulation and damages on cell viability. Alternatively, microfluidics offer the ability to localize the physical fields precisely to the cellular scale, holding great potential to boost exosome secretion while maintaining high cell viability.[2]

METHODS

As shown in Figure 1(a), a high-throughput microfluidic device is developed to mechanically squeeze cells to promote exosome secretion. Each device integrate a microchannel with 10 fishbone-shaped squeezing ridges fabricated by soft lithography on polydimethylsiloxane. The device layout is designed to repeatedly exert mechanical compression upon cells and reduce clogging issue.

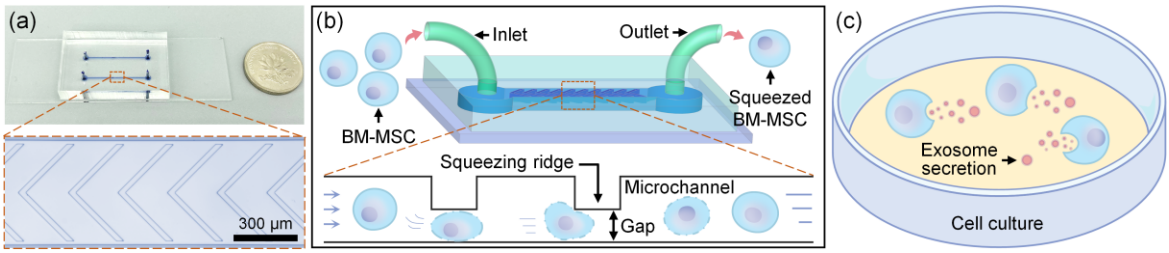

Figure 1: (a) Photograph of the device. (b) Schematic illustration on the working principle of the device. (c) Exosome secretion are highly improved from the cells treated by the microfluidic device.

The MSCs derived from human fetal bone marrow (BM-MSCs) with a concentration of 2×10^6 cells/mL are introduced into the device with ridge gap of 50% of the average cell diameter, while the flow rate is maintained at 32 μL/min (Figure 1(b)). As the cells rapidly pass through the microchannel, they are mechanically deformed repeatedly, resulting in transient membrane disruption and cell permeabilization. Thereafter, the device-treated BM-MSCs are collected at the outlet and re-cultured in culture flasks (i.e., the device group). During the cell culture, the permeabilization of cell membrane boosts the BM-MSCs to secrete exosomes (Figure 1(c)). As the

control group, the BM-MSCs not treated by the device are cultured under the same experimental condition to set the base level of secretion amount.

RESULTS

Transmission electron microscopy (TEM) images show that the exosomes derived from both untreated (i.e., the control group) and device-treated BM-MSCs (i.e., the device group) present a cup-shaped morphology (Figure 2(a)-(b)). Western blot (WB) detections show that both samples groups are positive for CD81, TSG101, and CD63 (Figure 2(c)), confirming their identity as exosomes. In addition, the nano-flow cytometry analysis reveal that these samples have a similar size range of 40 – 200 nm (Figure 2(d)). Moreover, each device-treated BM-MSC results in a secretion amount of exosomes 3.1-fold increase compared with the untreated cell (Figure 2(e)).

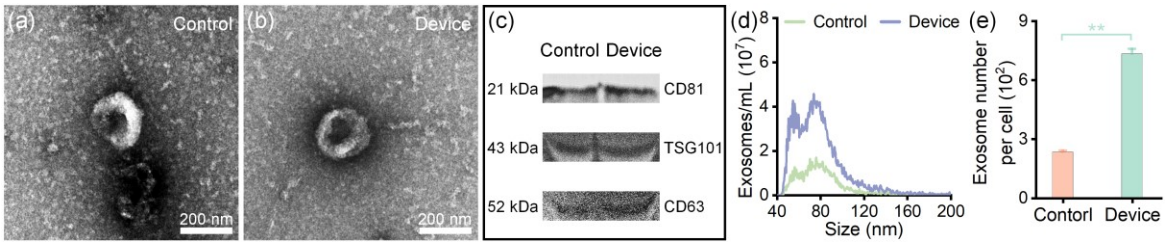

*Figure 2: (a)-(d) TEM images, WB detections, and size distributions of exosome samples derived from BM-MSCs. (e) Number of exosomes secreted by each cell. Data are presented as mean ± SD of three duplicated experiments (**P < 0.01).*

Further, exosome uptake by immortalized human corneal epithelial cells (iHCECs) is analyzed. Both groups of exosomes are labeled with fluorescent dye PKH26 and incubated with iHCECs. To eliminate background noise, iHCECs are also incubated with the same dye solution (i.e., the background group). As shown in Figure 3(a)-(c), the PKH26 is observed in the cells post-incubated with labeled exosomes, indicating exosomes derived from the treated BM-MSCs can be uptaken efficiently. Moreover, the average fluorescence intensity (FI) of PKH26 in each iHCEC of the device group is increased 2.9-fold compared to that from the control group (Figure 3(d)).

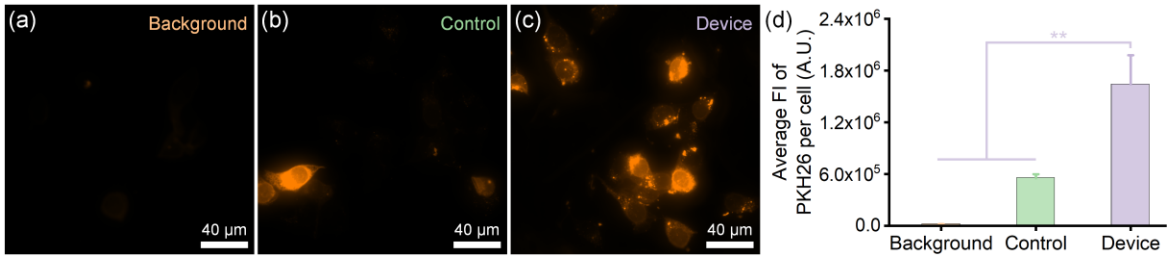

*Figure 3: (a)-(c) Fluorescence images on iHCECs co-incubated with PKH26 labeled samples. (d) Average FI of PKH26 per cell quantified from (a)-(c). Data are presented as mean ± SD of three duplicated experiments (**P < 0.01).*

CONCLUSION

In conclusion, we have presented a microfluidic device for promoting exosome secretion from BM-MSCs by mechanical stimulation. The microfluidic device works at a flow rate of 32 μL/min for 30 min without obvious clogging issue. Moreover, repeatedly compression acting on BM-MSCs can effectively permeabilize cell membrane, thereby improving exosome yield. Compared to the control group, exosomes secreted by the device-treated BM-MSCs is increased 3.1-fold. In addition, mechanical compression does not affect the bio-activity of exosomes. Hence, this platform could evolve into a robust tool to boost exosome secretion from various cell types.

ACKNOWLEDGEMENTS

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